

Transformation of Functions

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: (4, 2)

_____ The graph of $g(x) = f(x - 4)$ is the graph of f shifted which direction and how far? (e.g. “right 4”)

ANSWER: up 5

_____ The graph of $g(x) = f(x + 2)$ is the graph of f shifted which direction and how far?

ANSWER: 3

_____ The point (2, 5) lies on f . Find its image on $g(x) = f(x) + 4$.

ANSWER: down 3

_____ The graph of $g(x) = -f(x)$ is the graph of f reflected over which axis?

ANSWER: (2, 9) # _____ The point (2, 5) lies on f. Find its image on $g(x) = f(x - 1)$.	ANSWER: y-axis # _____ The graph of $g(x) = 3f(x)$ is a vertical stretch of f by what factor?
ANSWER: left 2 # _____ The graph of $g(x) = f(x) - 3$ is the graph of f shifted which direction and how far?	ANSWER: right 4 # _____ The graph of $g(x) = f(x) + 5$ is the graph of f shifted which direction and how far?
ANSWER: x-axis # _____ The graph of $g(x) = f(-x)$ is the graph of f reflected over which axis?	ANSWER: (3, 5) # _____ The point (4, 0) lies on f. Find its image on $g(x) = -f(x) + 2$.